

181. $2x + y = 5$...(i) and $3x - 4y = 2$...(ii)
 Multiplying (i) by 4 and adding (ii) to it, we get:
 $11x = 22$ or $x = 2$.
 Putting $x = 2$ in (i), we get: $y = 1$.
 So, $2xy = 2 \times 2 \times 1 = 4$.

182. $3x - 5y = 5$...(i) and $\frac{x}{x+y} = \frac{5}{7} \Rightarrow 7x = 5x + 5y$
 $\Rightarrow 2x - 5y = 0$...(ii)
 Subtracting (ii) from (i), we get: $x = 5$.
 Putting $x = 5$ in (i), we get: $y = 2$. So, $x - y = 5 - 2 = 3$.

183. $4x + 3y = 18xy$...(i) and $2x - 5y = -4xy$...(ii)
 Dividing (i) and (ii) by xy , we get:
 $\frac{3}{x} + \frac{4}{y} = 18$...(iii) and $\frac{5}{x} - \frac{2}{y} = 4$...(iv)

Multiplying (iv) by 2 and adding (iii) to it, we get:
 $\frac{13}{x} = 26$ or $x = \frac{1}{2}$.

Putting $x = \frac{1}{2}$ in (iii), we get: $y = \frac{1}{3}$.

184. $2x + y = 17$...(i)
 $y + 2z = 15$...(ii)
 and $x + y = 9$...(iii)
 Subtracting (iii) from (i), we get: $x = 8$.
 Putting $x = 8$ in (i), we get: $y = 1$. Putting $y = 1$ in (ii),
 we get: $2z = 14$ or $z = 7$.
 $\therefore 4x + 3y + z = 4 \times 8 + 3 \times 1 + 7 = 42$.

185. $3x - 4y + z = 7$...(i)
 $2x + 3y - z = 19$...(ii)
 and $x + 2y + 2z = 24$...(iii)
 Adding (i) and (ii), we get: $5x - y = 26$...(iv)
 Subtracting (i) from (ii) and adding to (iii), we get:
 $9y = 36$ or $y = 4$.
 Putting $y = 4$ in (iv), we get: $5x = 30$ or $x = 6$.
 Putting $x = 6, y = 4$ in (iii), we get: $2z = 10$ or $z = 5$.

186. $2x + y = 15$...(i)
 $2y + z = 25$...(ii)
 and $2z + x = 26$...(iii)
 Adding (i), (ii) and (iii), we get: $3(x + y + z) = 66$ or $x + y + z = 22$...(iv)
 From (ii), we have: $y = \frac{25 - z}{2}$. From (iii), we have:
 $x = 26 - 2z$.

$$\therefore (26 - 2z) + \left(\frac{25 - z}{2}\right) + z = 22 \Leftrightarrow 77 - 3z = 44$$

$$\Leftrightarrow 3z = 33 \Leftrightarrow z = 11.$$

187. $2x + 3y = 31$...(i)
 $y - z = 4$...(ii)
 and $x + 2z = 11$...(iii)
 Multiplying (iii) by 2 and subtracting from (i), we get:
 $3y - 4z = 9$...(iv)
 Solving (ii) and (iv), we get: $y = 7, z = 3$.
 Putting $y = 7$ in (i), we get: $x = 5$.
 $\therefore x + y + z = (5 + 7 + 3) = 15$.

188. Let the cost of a chair and that of a table be ₹ x and ₹ y respectively.

$$\text{Then, } 10x = 4y \text{ or } y = \frac{5}{2}x.$$

$$\therefore 15x + 2y = 4000 \Leftrightarrow 15x + 2 \times \frac{5}{2}x = 4000 \Leftrightarrow 20x = 4000 \Leftrightarrow x = 200.$$

$$\text{So, } y = \left(\frac{5}{2} \times 200\right) = 500.$$

$$\therefore \text{Cost of 12 chairs and 3 tables} = 12x + 3y \\ = ₹ (12 \times 200 + 3 \times 500) = ₹ (2400 + 1500) \\ = ₹ 3900.$$

189. Let the price of a jean be ₹ x and that of a shirt be ₹ y .
 Then, $2x + 3y = 4000$...(i) and $3x + 2y = 3500$...(ii)
 Adding (i) and (ii),
 we get: $5x + 5y = 7500$ or $x + y = 1500$...(iii)
 Subtracting (ii) from (i), we get: $-x + y = 500$...(iv)
 Adding (iii) and (iv), we get: $2y = 2000$ or $y = 1000$.
 Putting $y = 1000$ in (iii), we get: $x = 500$.
 Hence, cost of a jean = ₹ 500.

190. Let the cost of one pen be ₹ x and that of one pencil be ₹ y .
 Then, $8x + 4y = 176$ or $2x + y = 44$...(i)
 And, $2x + 2y = 48$ or $x + y = 24$...(ii)
 Subtracting (ii) from (i), we get: $x = 20$.
 Hence, cost of 1 pen = ₹ 20.

191. Let the cost of 1 dozen apples be ₹ x and that of 1 dozen bananas be ₹ y . Then,
 $2x + 3y = 136$...(i) and, $x + 5y = 110$ or $2x + 10y = 220$...(ii)
 Subtracting (i) from (ii), we get: $7y = 84$ or $y = 12$.

192. Let the cost of 1 pencil, 1 pen and 1 eraser be ₹ x , ₹ y and ₹ z respectively.
 Then, $x + 2y + 4z = 22$...(i)
 and $5x + 4y + 2z = 32$...(ii)
 Adding (i) and (ii), we get:
 $6(x + y + z) = 54$ or $x + y + z = 9$.
 $\therefore$ Cost of 3 pencils, 3 pens and 3 erasers
 $= 3(x + y + z) = ₹ (3 \times 9) = ₹ 27$.

193. Let the number of cows and horses bought be x and y respectively.
 Then, number of chicken bought = $100 - (x + y)$.
 $1000x + 300y + 50 [100 - (x + y)] = 10000$
 $\Rightarrow 1000x + 300y + 5000 - 50(x + y) = 10000$
 $\Rightarrow 950x + 250y = 5000 \Rightarrow 50(19x + 5y) = 5000$
 $\Rightarrow 19x + 5y = 100 \Rightarrow 5y = 100 - 19x \Rightarrow y = \frac{100 - 19x}{5}$.
 Clearly, for y to be a whole number, $(100 - 19x)$ must be divisible by 5 and this is possible only when $x = 5$.
 When $x = 5, y = \frac{100 - 19 \times 5}{5} = \frac{5}{5} = 1$.
 $\therefore$ Number of chicken = $[100 - (5 + 1)] = 94$.